DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

Academic Year: 2025-26	**Year:** Second Year	**Semester:** II
PRN No.: 1012411011	**Name:** Riddhee Kulkarni	
Subject: Python Programming		
Assignment No.: 26		
Date: 08/04/2026		

Lab Assignment: 26

Aim:
To write a program that reads data from a file and calculates the percentage of vowels and consonants in the file.

Objective:
- To understand file handling and reading data from a file in Python.
- To identify and differentiate vowels and consonants in text.
- To calculate percentages based on character frequency.
- To enhance string processing and analytical skills.

Theory:
File handling in Python allows reading data from files using functions like open() and read(). Once the file content is read, it is treated as a string and can be processed using various string operations.

In this program, the characters in the file are analyzed to count vowels and consonants. Vowels include the letters **a, e, i, o, u** (both uppercase and lowercase), while consonants are all other alphabetic characters except vowels.

The program iterates through each character of the file content, checks whether it is a vowel or consonant, and increments the respective counters. Non-alphabetic characters such as digits, spaces, and symbols are ignored.

Finally, the percentage of vowels and consonants is calculated using the formula:
Percentage = (Count / Total Alphabets) × 100

This helps in understanding text composition and applying basic data analysis on file content.
.

Algorithm:

1. Start the program.
2. Accept the filename from the user.
3. Open the file in read mode.
4. Read the content of the file.
5. Initialize counters for vowels and consonants.
6. Traverse each character in the file content.
7. Check if the character is an alphabet.
8. If it is a vowel, increment vowel count.
9. Else, increment consonant count.
10. Calculate total number of alphabets.
11. Calculate percentage of vowels and consonants.
12. Display the results.
13. Close the file.
14. End the program.

Pseudo Code:

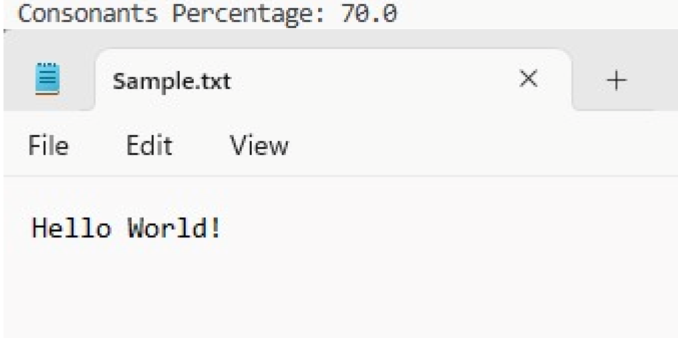
```
START
INPUT filename
OPEN file in read mode
READ file content
SET vowel_count = 0
SET consonant_count = 0
FOR each character in content
 IF character is alphabet
  IF character is vowel
   INCREMENT vowel_count
  ELSE
   INCREMENT consonant_count
 END IF
END FOR
SET total = vowel_count + consonant_count
CALCULATE vowel percentage
CALCULATE consonant percentage
PRINT vowel percentage
PRINT consonant percentage
CLOSE file
END
```

Code:

```python
# Take filename from user
filename = input("Enter file name: ")
try:
    file = open(filename, "r")
    content = file.read().lower()   # convert to lowercase
    vowels = "aeiou"
    v_count = 0
    c_count = 0
    for ch in content:
        if ch.isalpha():   # consider only letters
            if ch in vowels:
                v_count += 1
            else:
                c_count += 1
    total = v_count + c_count
    if total > 0:
        v_percent = (v_count / total) * 100
        c_percent = (c_count / total) * 100

        print("Vowels Percentage:", v_percent)
        print("Consonants Percentage:", c_percent)
    else:
        print("No alphabets found in file")

    file.close()

except FileNotFoundError:
    print("File not found")
```

Output:

```
Enter file name: Sample.txt
Vowels Percentage: 30.0
Consonants Percentage: 70.0
```



Conclusion:

Thus, the Python program was successfully executed to read data from a file and calculate the percentage of vowels and consonants. This demonstrated effective use of file handling, character analysis, and basic percentage calculations in Python.